

Final Lab – Quanser Aero Control



AE-443 Controls Lab / Prof. Restrepo-Torres

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I. ABSTRACT

In this report, PID controllers of the Quanser Aero system were modified using trial and error runs and theory analysis. For the open loop test, measured values resulted in 1.528s for Peak Time, 75% Overshoot and 7.56s for Settling Time. Iteratively changing the proportional control with $K_p = 5, 7.5, 10$ resulted in faster response time, but unstable settling; however, iterative derivative control testing with $K_d = 4, 6, 8$ reduced the overshoot. Changing the integral control with $K_i = 3, 4.5, 6$ reduced the steady-state error, showing that a balance between all three would allow the system to be stable and respond appropriately. A tuned PID controller using $K_p = 15$, $K_d = 4.5$, and $K_i = 5$ closely modeled the reference input and met all three specifications for peak pitch, peak time, and steady-state error. The Aero was also modeled as a second-order system with a natural frequency of 2.3463 rad/s and a damping ratio of 0.0675, which was used to analytically calculate PID gains of $K_p = 3.46$, $K_i = 3.17$, and $K_d = 3.88$. These gains met the overshoot and steady-state specifications but produced a peak time of 3.15 seconds, exceeding the 1.1 second requirement, reflecting the fundamental difference between a critically damped design and one tuned for fast response.

II. PROCEDURE

The purpose of this experiment is to implement the concepts of the proportional, integral, and derivative (PID) controller into the Quasar motor system.

The Simulink diagram: q_aero_PID_specification.mdl is used for the entire PID experiment as it applies the command and unity feedback for the motors. The initial trial will analyze an open-loop response of the system to later compare the results of integrating a full PID controller into it. The step input is then set to an initial value of 10 and a final value of 15. After executing the script to the Aero, the system response will be recorded and plotted to a pitch angle response (radians) vs time (seconds) graph. The data from the graph is then used to measure and calculate the following variables: Peak time, settling time within 5%, frequency, damping, and percent overshoot.

II.1 Proportional Control

Proportional control is determined by the difference between the desired response of a system and its measured response [1]. After running the open-loop response, a proportional control (K_p) block was added between the error signal and the motor voltage input in the diagram. The model ran several times, with each response used varying K_p values between 5 and 10. Every response generated from the proportional values were plotted onto the same graph be observed and assessed to understand the effect of a proportional controller on a system's peak time, oscillations, and overshoot.

II.2 Derivative Control

The derivative control (K_d) was inserted into the diagram to reduce system overshoot and improve stability [1]. The K_d gain block along with its transfer function was placed in parallel with the proportional controller block to allow for the summation of both controllers. With the proportional gain block set to 10, the derivative control block was set with varying quantities between 4 and 8. Afterwards, the system input was switched from step input to square wave input, directing the focus towards lessening the system overshoot and increasing stability. The PD model was then reconfigured with a parameter of a 0.3 radian square wave and analyzed different values of the derivative controller gain. The gain was determined in order to satisfy the condition of producing a peak pitch that stayed within ± 0.35 radians [1].

II.3 Integral Control

Integral control (K_i) was used to ensure the system would settle close to the setpoint which would otherwise not occur with purely proportional and derivative control [1]. By adding an integral compensator, we could track the integral of the error to make changes to the system. An integrator block with integral gain k_i was placed in parallel with the two gains which can be found within the documentation in figure 2.5. The proportional and derivative gain values were

kept the same as before, and the integral gain was varied in steps between 3 and 6 to determine the effects on the system response.

II.4 Response Tuning

All three controller gains for the Aero were finely altered and tuned to meet within the constraints of the lab specifications: A peak time of less than 1.1 seconds, a peak pitch that deviates less than 0.35 radians, with a negligible steady-state error. The response of the modified system is plotted along with the input signal for comparison, allowing to determine if the gains meet the requirements of the controller.

II.5 Second Order System Approximation

Within the context of the lab, the Quanser Aero motor system is modeled as a second-order system to calculate and determine the proportional, integral, and derivative gains that meet the system's performance criteria.

The PID controller was modified by resetting the control gain values: $K_p = 1$, $K_i = K_d = 0$. The changes for the controller enables a unity feedback loop. With the step input activated and its gain set to 1, the response was recorded and plotted. The data from the response plot was extracted to determine the peak time (t_p), peak response (y_{max}), steady-state value (y_f), and the peak overshoot.

The percent overshoot (PO) was determined through the equation:

$$PO = 100 \left(\frac{y_{max} - y_f}{y_f} \right) \quad (2.1)$$

where the damping ratio ζ is calculated through the relationship with the percent overshoot:

$$PO = 100e^{\left(\frac{-\pi\zeta}{\sqrt{1-\zeta^2}} \right)} \quad (2.2)$$

The peak time was calculated through the formula with the damping ratio:

$$t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} \quad (2.3)$$

From there, a second-order transfer function was modeled to approximate system behavior.

II. Calculating PID Gains

After acquiring the second-order system, the gains from the PID were calculated from the standard controller transfer function:

$$C(s) = K_p + \frac{K_i}{s} + K_d s \quad (2.4)$$

Each of the controller gains were determined through derived methods of the relationships involving the natural frequency ω_n , damping ratio ζ , and zero position p_0 . The Aero system was

tested again but with different parameters. The design criteria was a critically damped system ($\zeta = 1$) with a settling time of 5% within a time constraint of 2 seconds. The input response for the trial is switched to square wave with the command gain set to 0.3. Following execution of the system, the responses are then plotted in a pitch angle (radians) vs time (seconds graph). The data of the graph is analyzed to conclude if the system was able to meet within the constraints for both damping and settling [1].

III. RESULTS

III.1. Open Loop Response

The open-loop response of the system was recorded with a step input configured from an initial value of 10 to a final value of 15. The angle and voltage responses are shown in Figures 1 and 2, and the measured system parameters are summarized in Table 1.

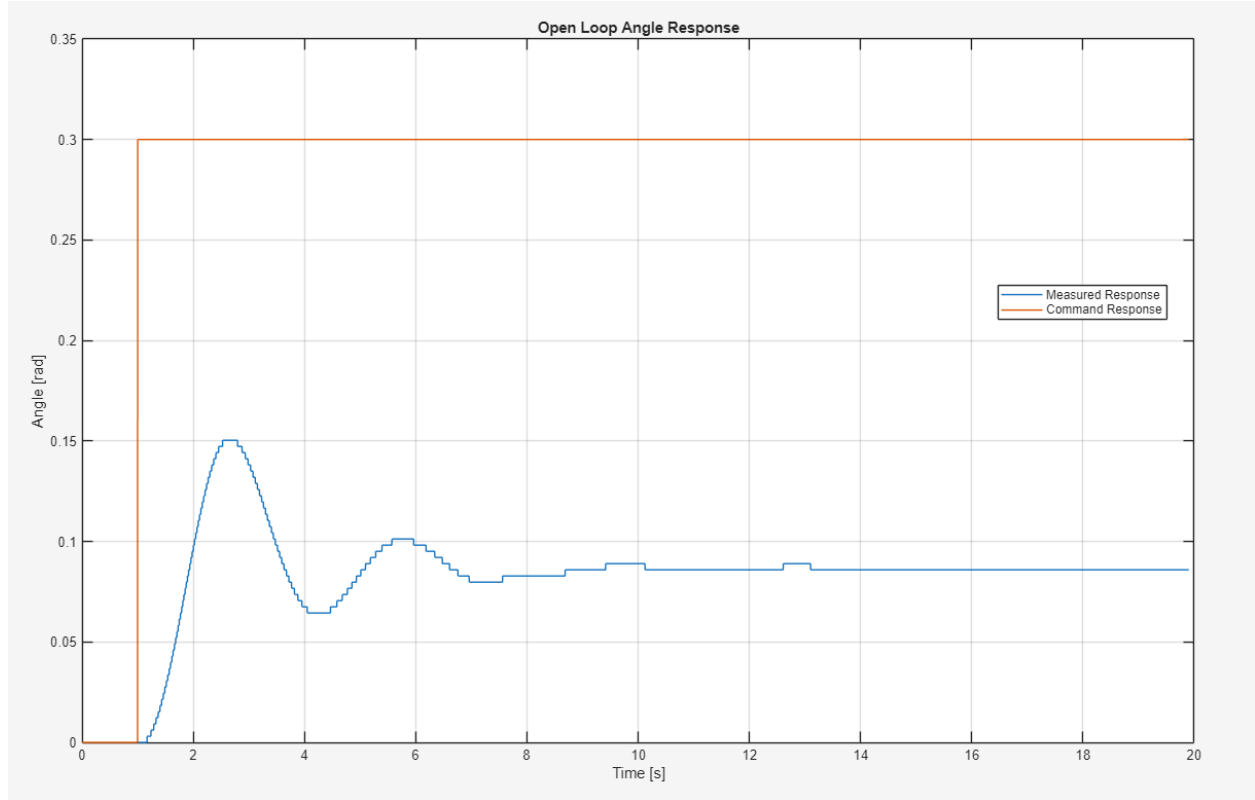


Figure 1: Open-loop pitch angle response to a step voltage input.

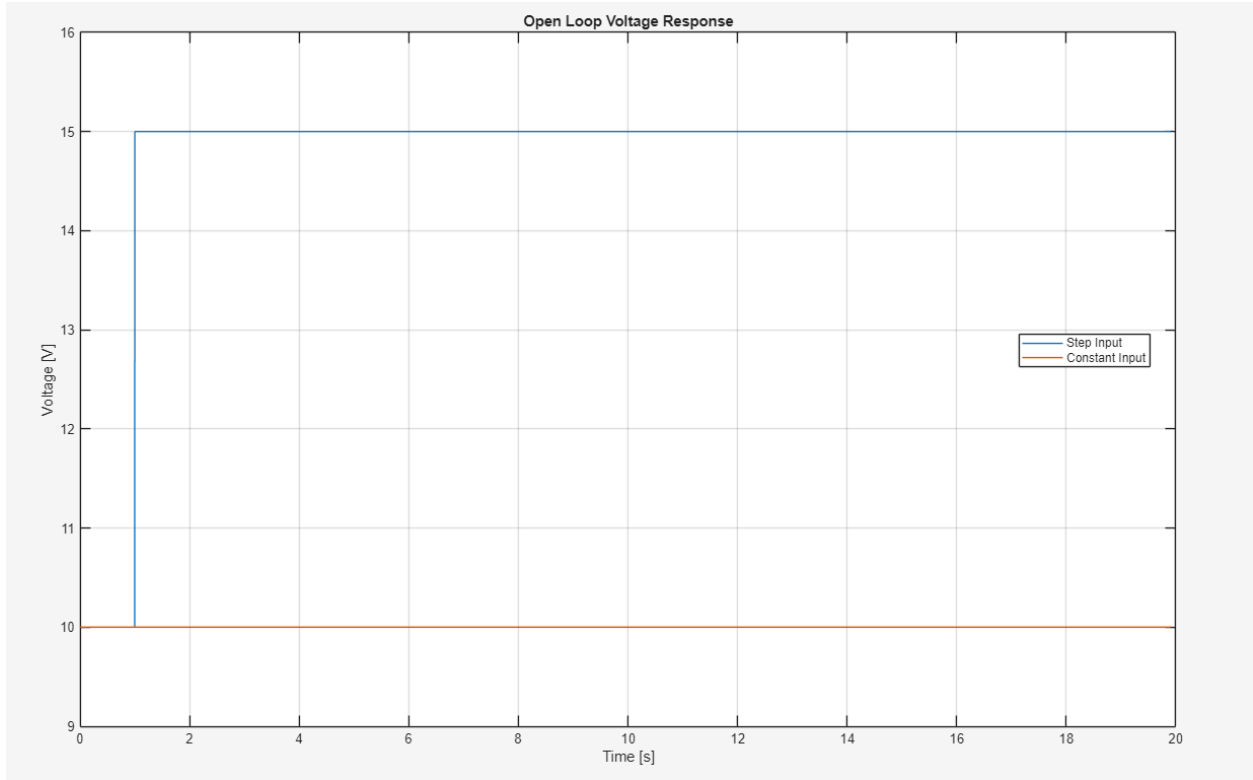


Figure 2: Open-loop motor voltage input showing the step from 10V to 15V.

Table 1: Open-loop response parameters measured from Figure 1.

Peak Time	Settling Time	Percent Overshoot	Damping Ratio	Natural Frequency	Damped Frequency
1.528 s	7.56 s	75.0%	0.0912	2.065 rad/s	2.056 rad/s

III.2 Proportional Control

With the proportional gain block added between the error signal and the motor voltage input, the system was run with proportional gain (K_p) values of 5, 7.5, and 10. Figure 3 shows the angle response for each gain alongside the command signal. Figure 4 shows the corresponding voltage responses for each gain value.

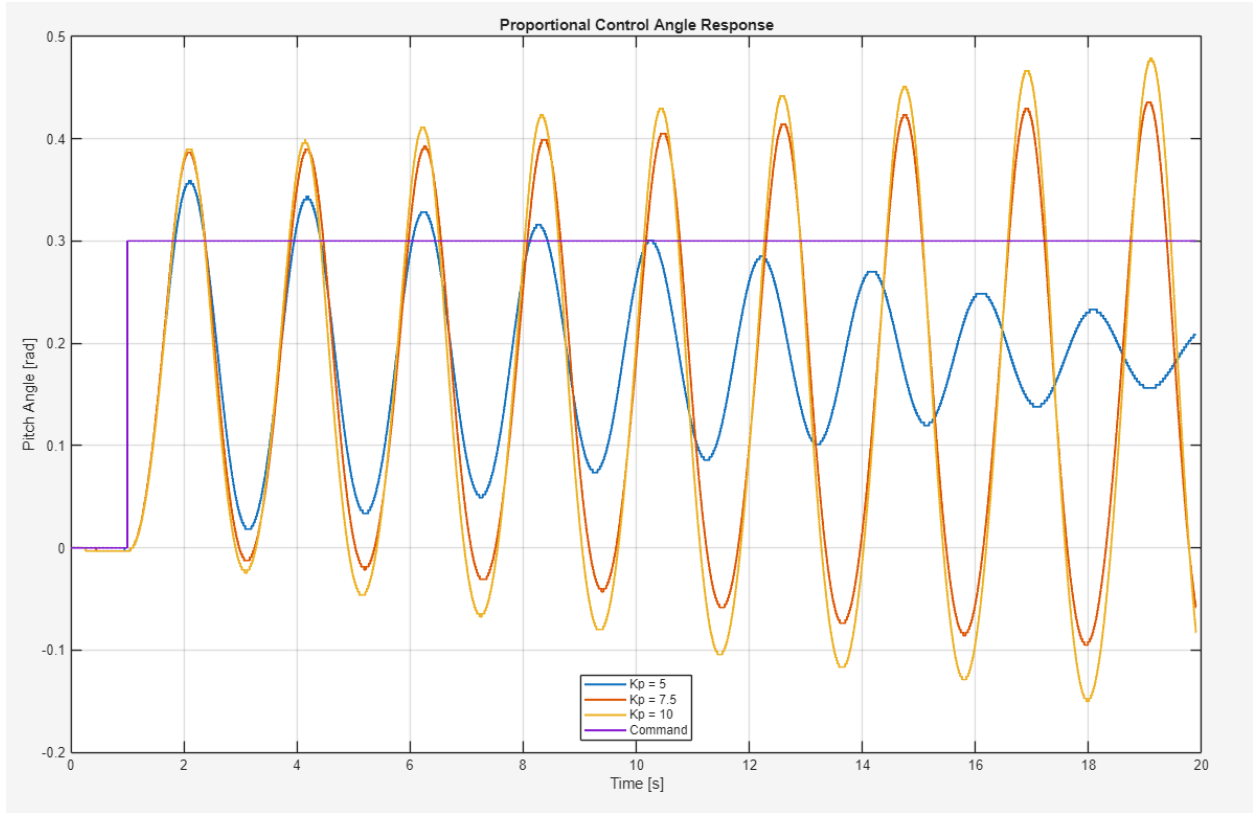


Figure 3: Proportional control pitch angle response for $K_p = 5, 7.5$, and 10 .

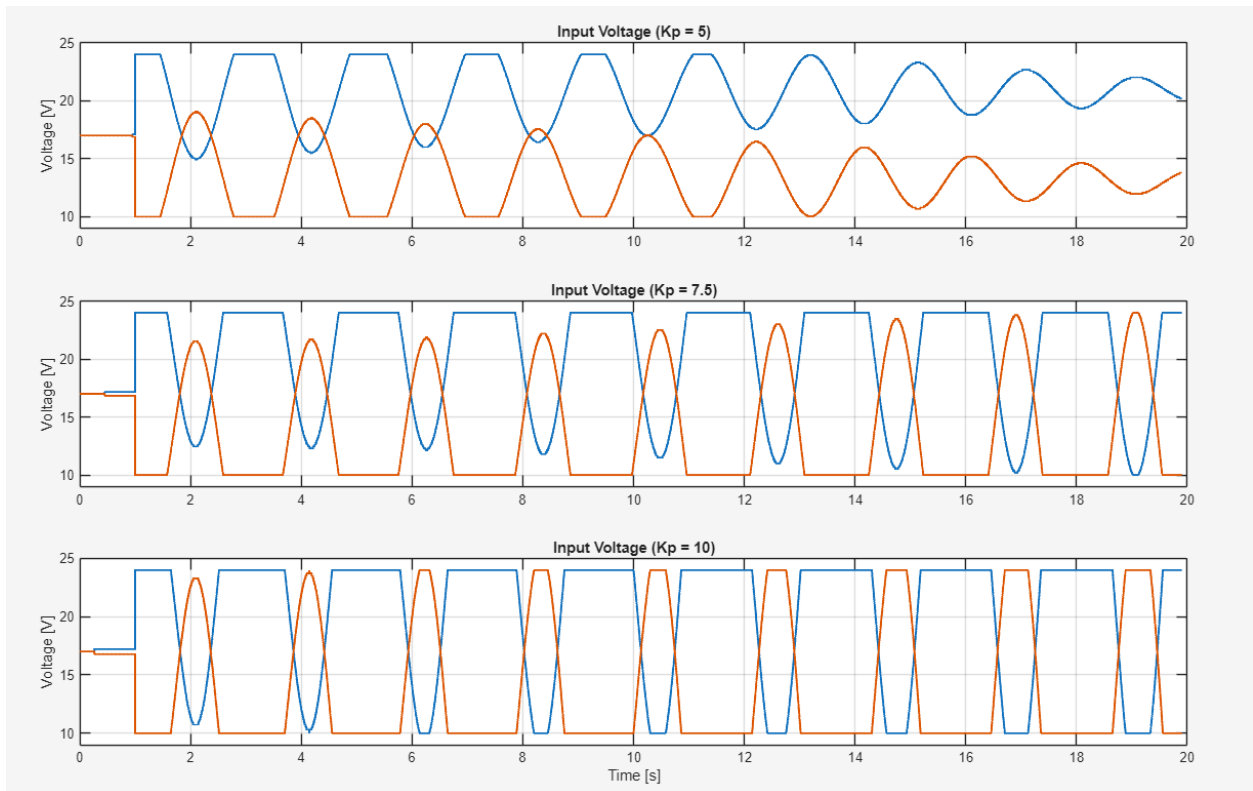


Figure 4: Input voltage response for $K_p = 5, 7.5$, and 10 .

III.3 Derivative Control

A derivative gain block was added in parallel with the proportional gain, with K_p fixed at 10. The system was tested with K_d values of 4, 6, and 8 using a step input, and then switched to a 0.3 radian square wave to evaluate overshoot performance. Figure 5 shows the angle responses for the three derivative gains. Figure 6 shows the corresponding voltage responses. Figure 7 shows the angle response for a small derivative gain of $K_d = 0.24$, demonstrating the instability and persistent oscillation that result when the derivative gain is too low. Figure 8 shows the tuned response with $K_d = 2.4$ using the square wave input, which kept the peak pitch within the ± 0.35 radian overshoot limit.

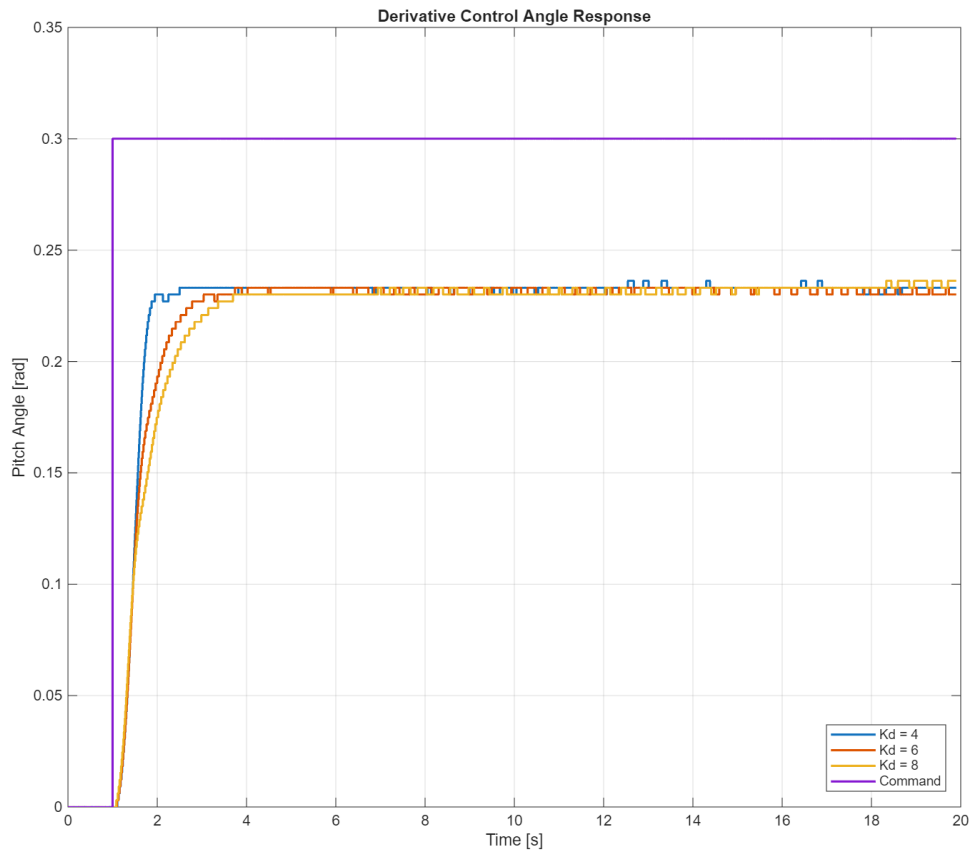


Figure 5: Derivative control pitch angle response for $K_d = 4, 6$, and 8 .

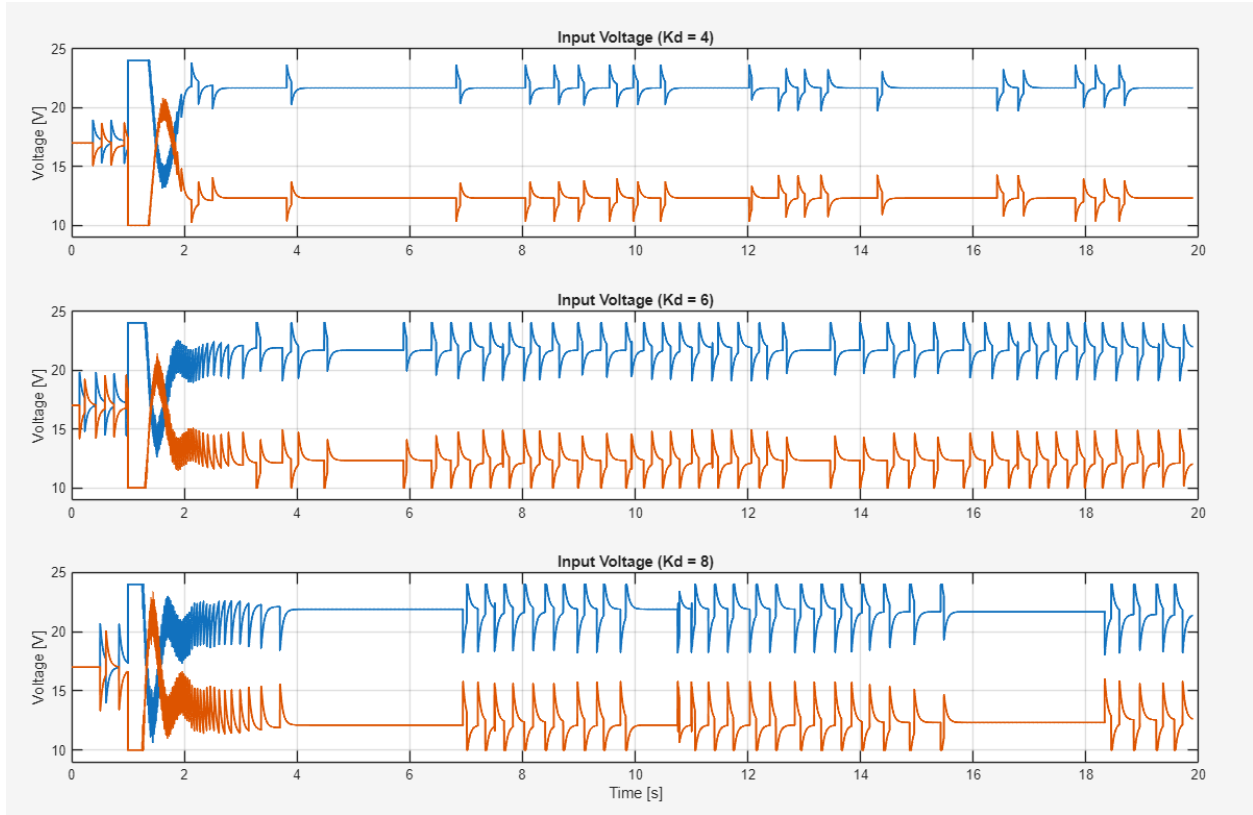


Figure 6: Input voltage responses for $K_d = 4, 6$, and 8 .

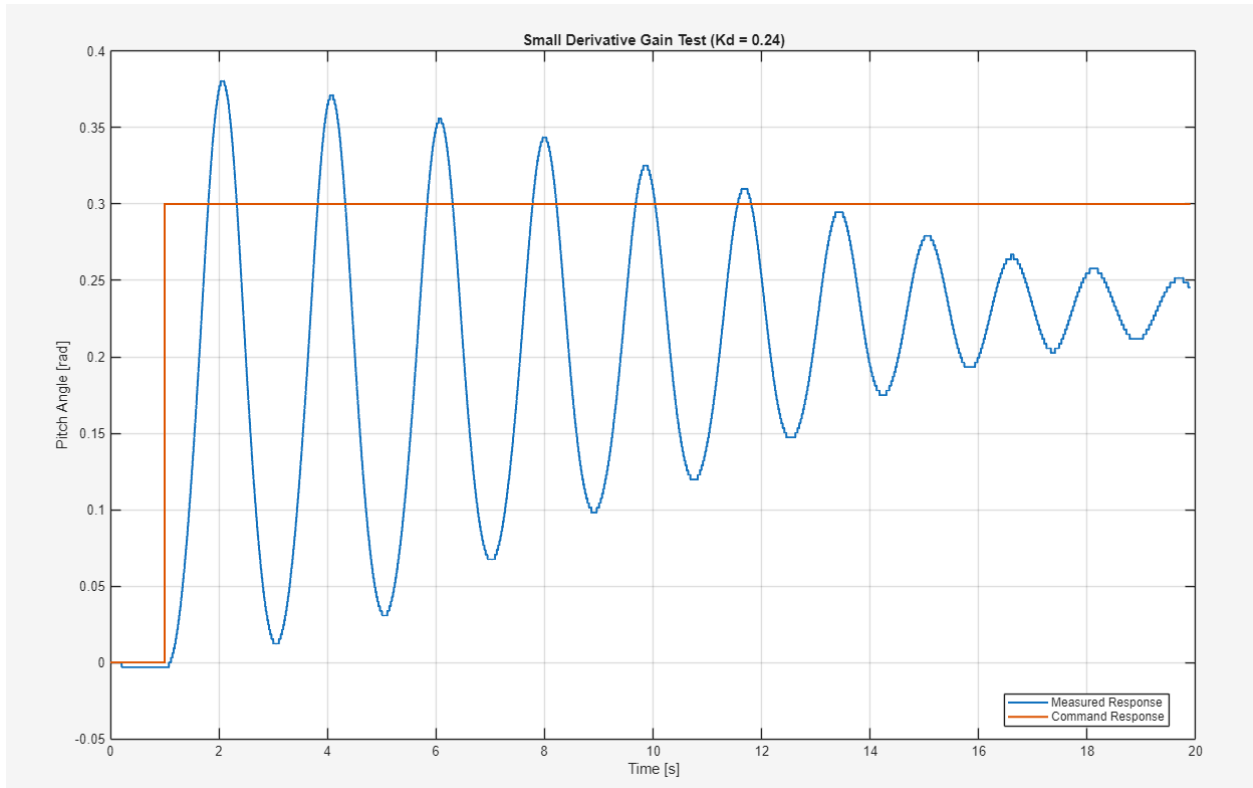


Figure 7: Pitch angle response for small derivative gain ($K_d = 0.24$).

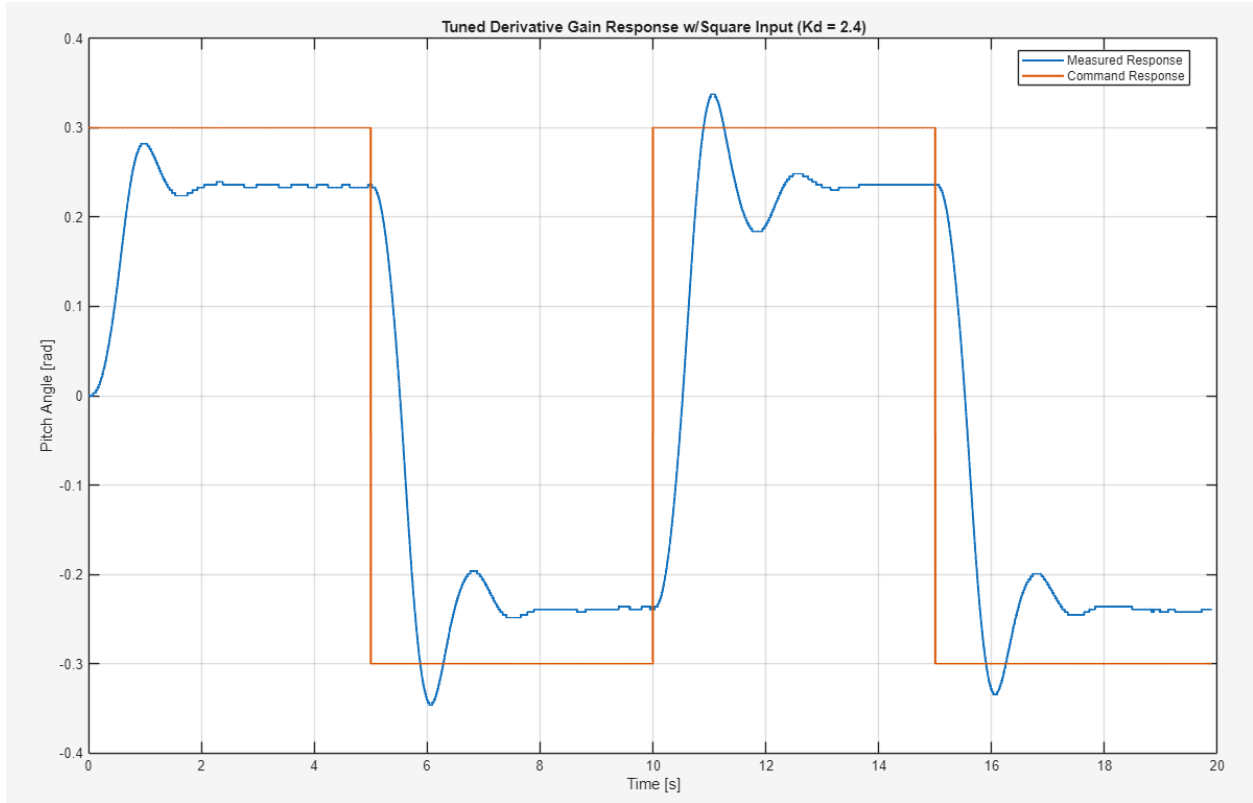


Figure 8: Tuned derivative gain response ($K_d = 2.4$) with square wave input.

III.4 Integral Control

With K_p and K_d held at the values identified in the derivative tuning step (10 & 2.4), an integral gain block was added to reduce the remaining steady-state error and pull the measured response closer to the commanded value. The system was tested with K_i values of 3, 4.5, and 6 using the 0.3 radian square wave input. Figure 9 shows the angle responses for all three integral gains overlaid with the command signal. Figure 10 shows the corresponding input voltage responses. Lastly, $K_i = 5$ was identified as the value that brings the steady-state response within the required 3% band of 0.291 to 0.309 radians, as shown in Figure 11.

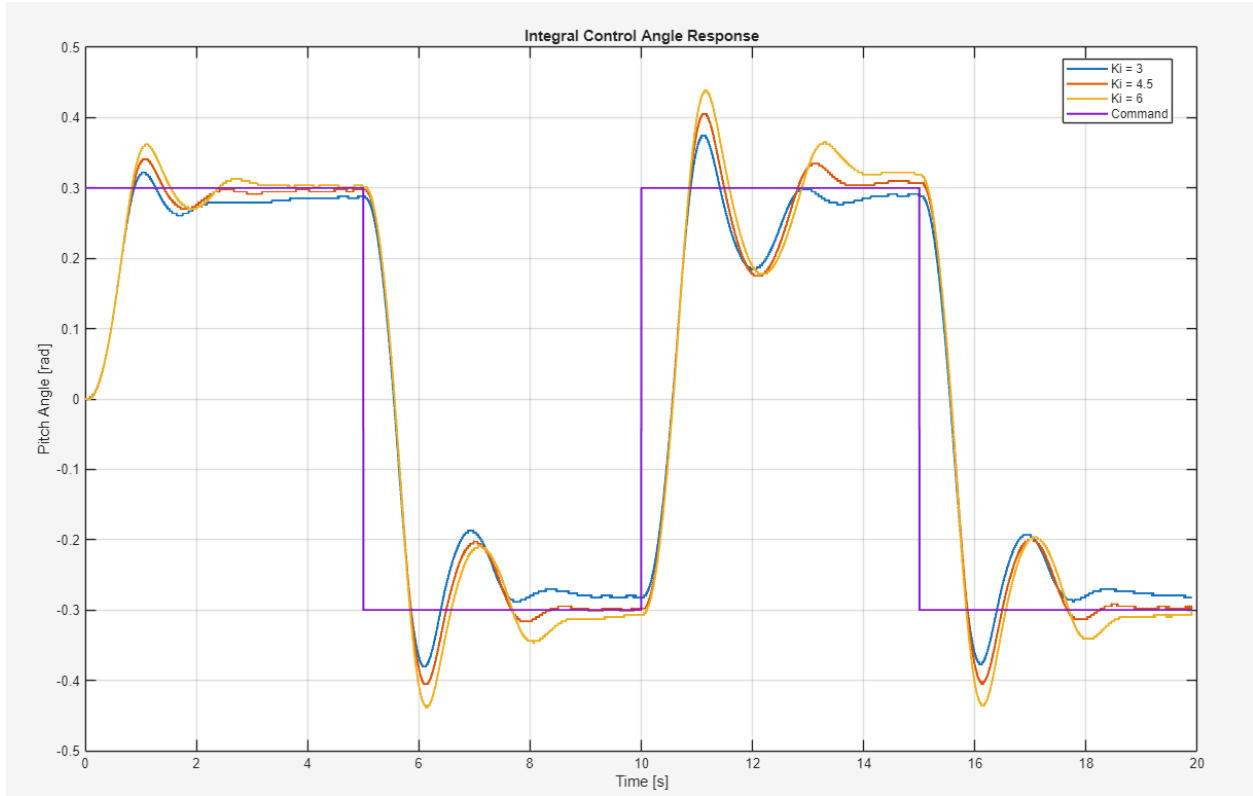


Figure 9: Integral control pitch angle response for $K_i = 3, 4.5$, and 6.

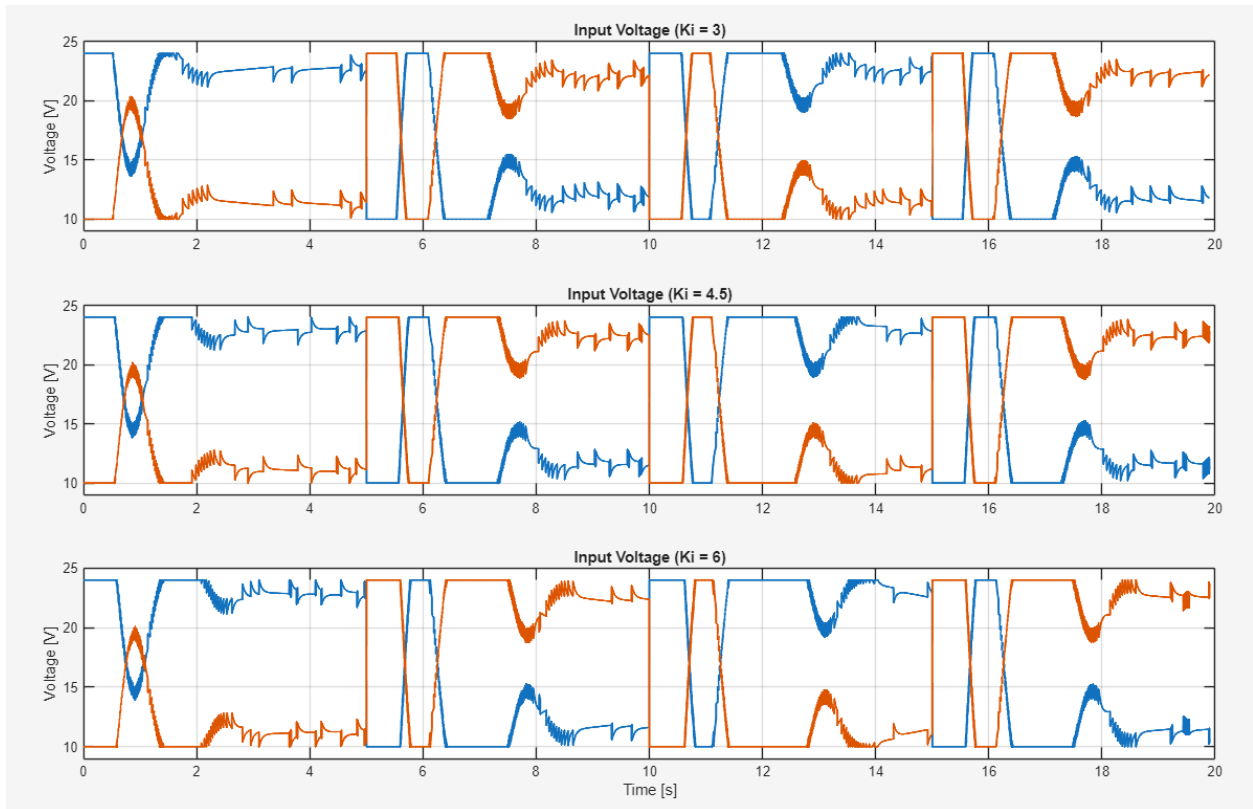


Figure 10: Input voltage responses for $K_i = 3, 4.5$, and 6.

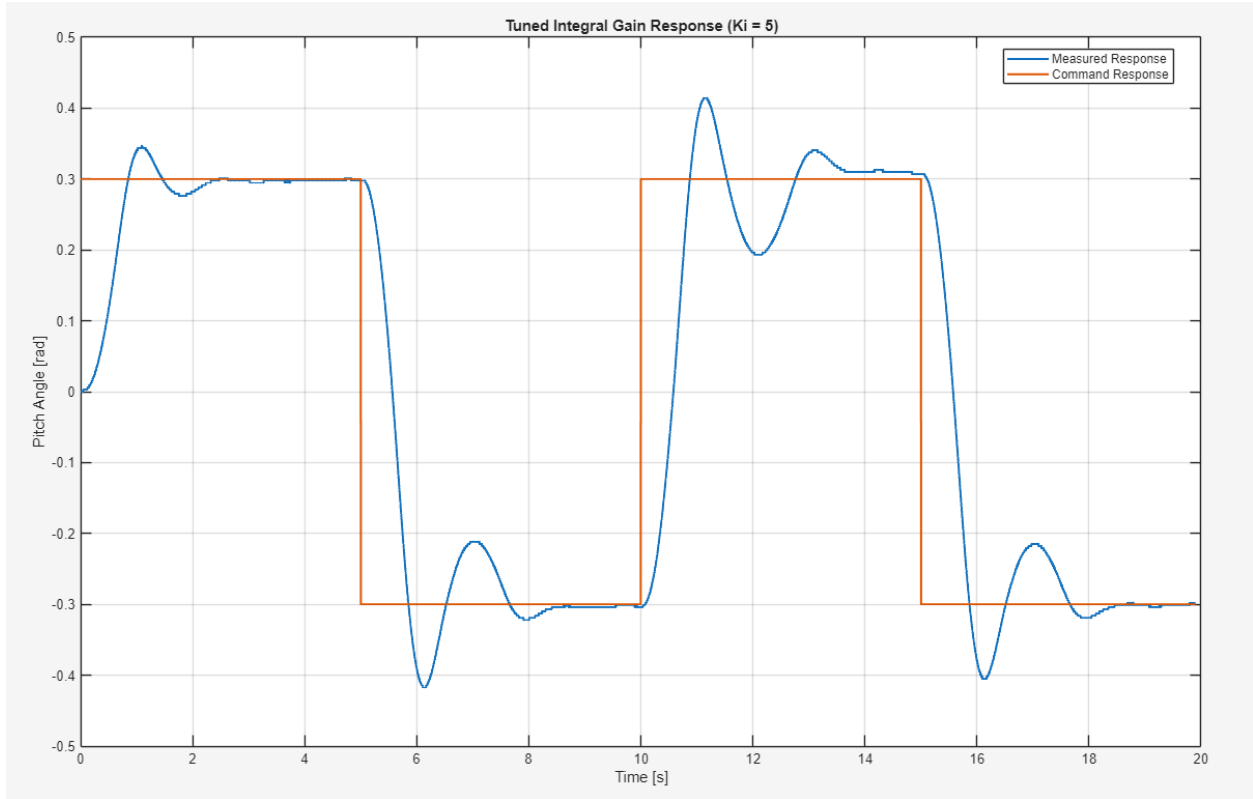


Figure 11: Tuned integral gain pitch angle response ($K_i = 5$).

III.5 Response Tuning

The PID gains were tuned by adjusting K_p , K_d , and K_i and testing the system response after each change until all three specifications were met. Figure 12 shows the tuned angle response alongside the command signal and overshoot limits, and Figure 13 shows the corresponding voltage response. The final tuned gains and measured response parameters are summarized in Table 2.

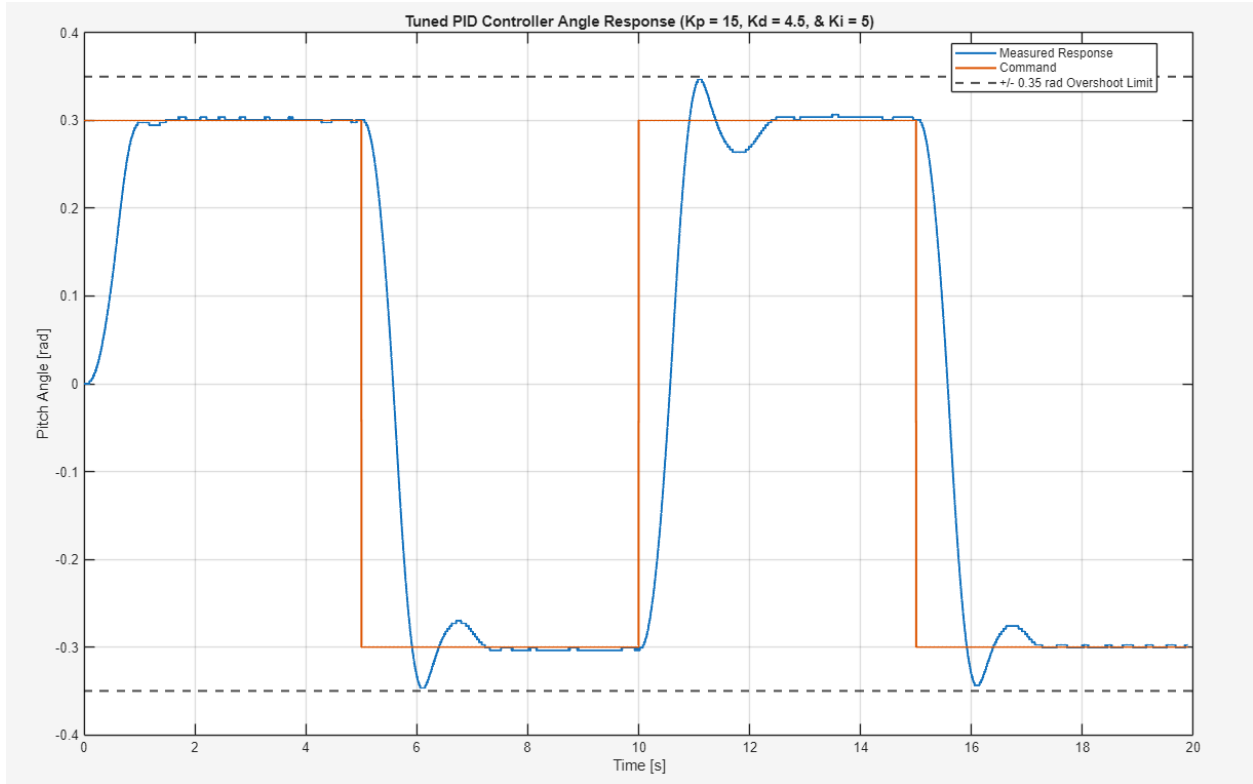


Figure 12: Tuned PID controller pitch angle response ($K_p = 15$, $K_d = 4.5$, $K_i = 5$).

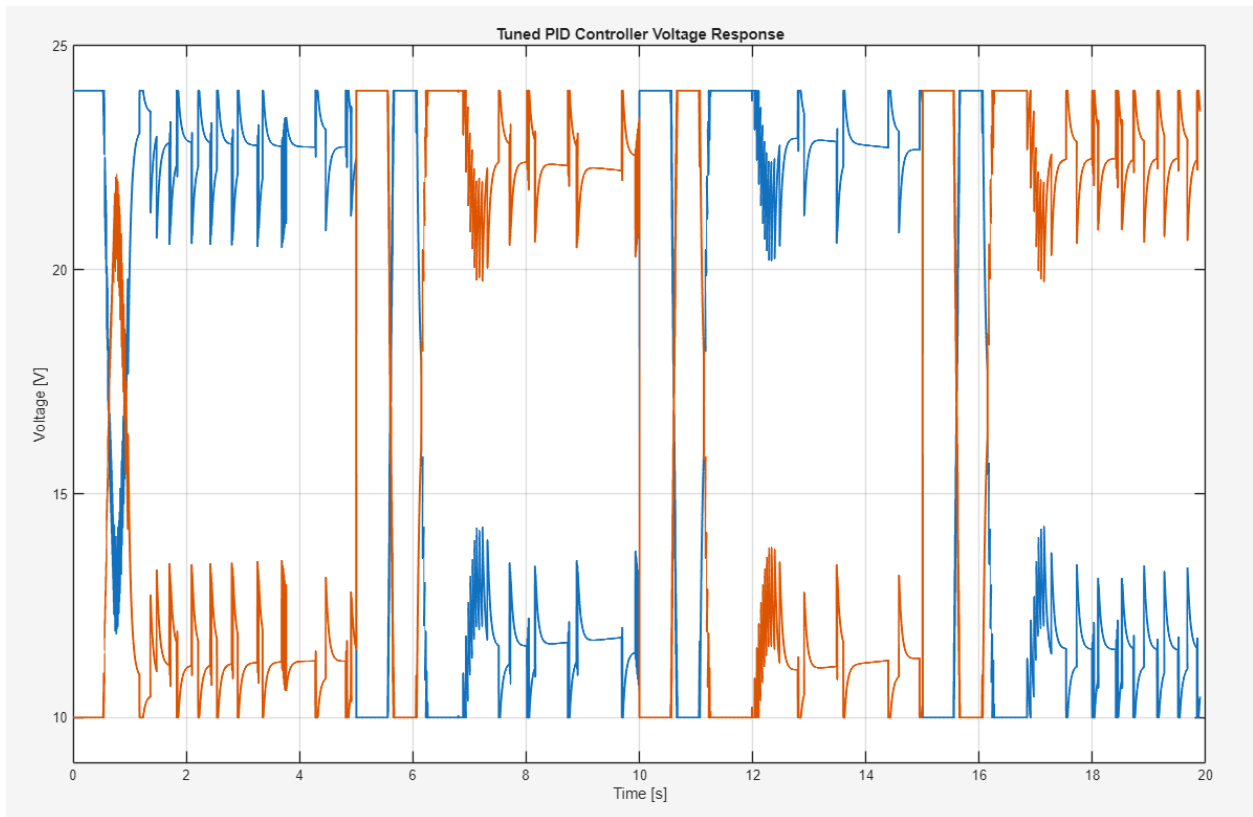


Figure 13: Tuned PID controller voltage response.

Table 2: Tuned PID Controller Response Parameters

Section	Description	Symbol	Value	Unit
Section 2.3	Derivative Tuning			
	Derivative Gain	K_d	2.4	-
	Peak Pitch	θ_{peak}	0.3375	rad
Section 2.4	Integral Tuning			
	Integral Gain	K_i	5	-
	Steady-State Value	θ_{ss}	0.2991	rad
Section 2.5	Response Tuning			
	Proportional Gain	K_p	15	-
	Integral Gain	K_i	5	-
	Derivative Gain	K_d	4.5	-
	Percent Overshoot	PO	15.56	%
	Peak Pitch	θ_{peak}	0.3467	rad
	Peak Time	t_{peak}	1.066	s
	Steady-State Value	θ_{ss}	0.3005	rad

III.6 Second Order System Approximation

The Quanser Aero was then configured with a unity feedback loop ($K_p = 1$, $K_d = 0$, $K_i = 0$) and subjected to a unit step input to define the dynamic behavior. Figure 14 shows the measured pitch angle response. The response was analyzed to extract t_p , y_{max} , and y_f , which were used to calculate the second-order system parameters ω_n and ζ , summarized in Table 3. Figure 15 shows the second-order transfer function model response overlaid with the measured Aero response, demonstrating how well the approximation captures the system's behavior.

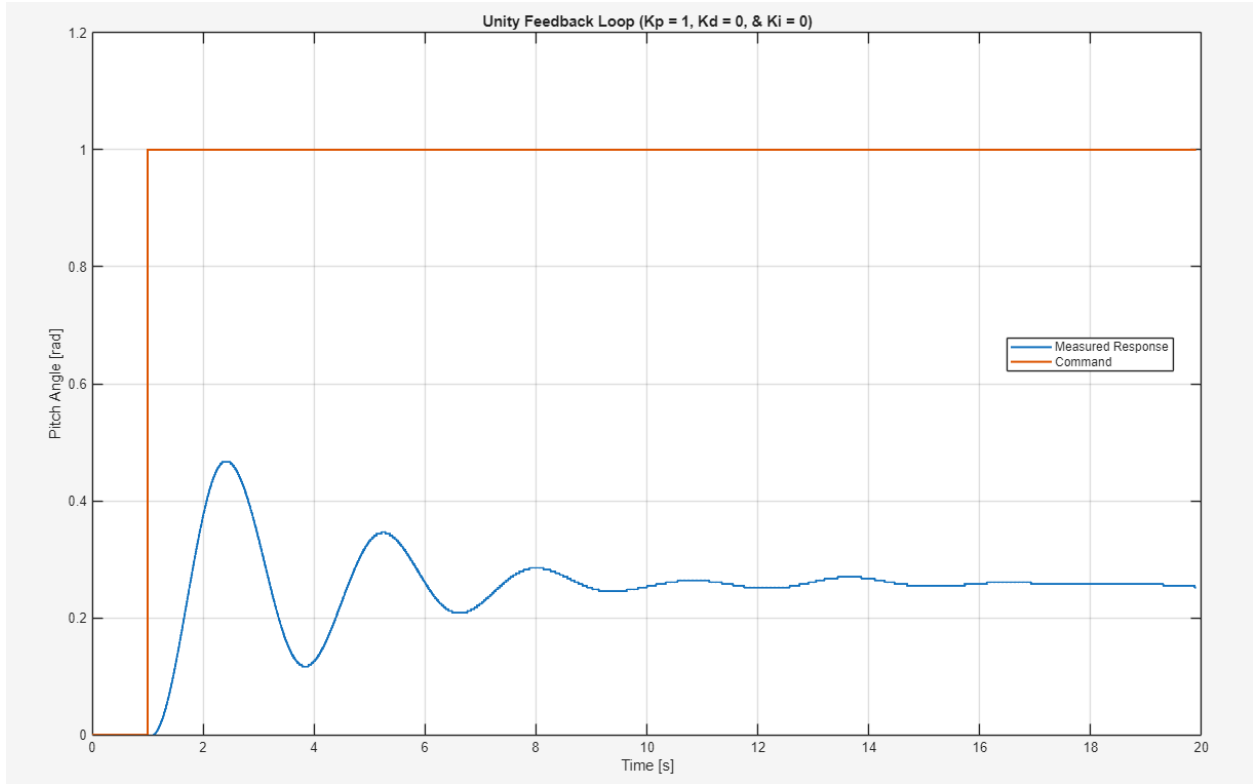


Figure 14: Unity feedback pitch angle response to a unit step ($K_p = 1$, $K_d = 0$, $K_i = 0$).

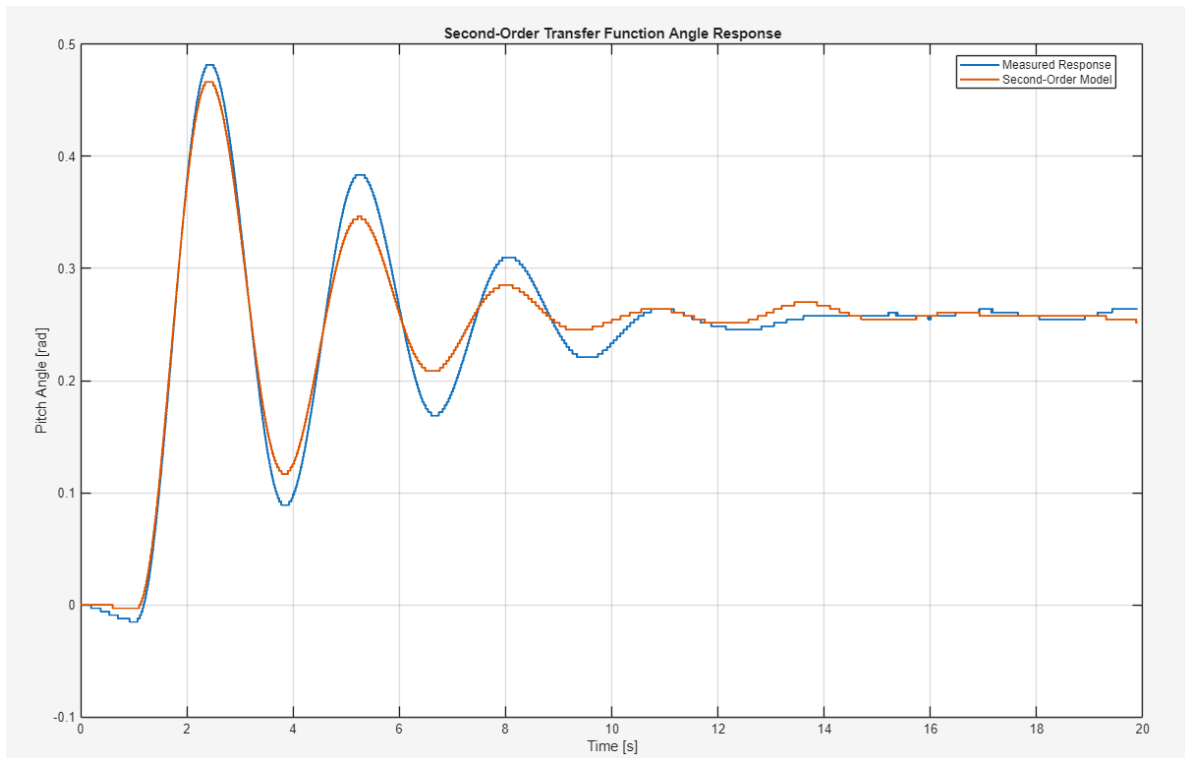


Figure 15: Comparison between the measured response and the second-order transfer function approximation.

Table 3: Second-Order System Approximation Parameters

Section	Description	Symbol	Value	Unit
Section 4.1.4	Peak Time	t_{peak}	1.342	s
	Peak Pitch	θ_{peak}	0.4663	rad
	Final Pitch	θ_f	0.2578	rad
Section 4.1.5	Natural Frequency	ω_n	2.3463	rad/s
	Damping Ratio	ζ	0.0675	-

The natural frequency and damping ratio were calculated by rearranging Equation 2.2 and Equation 2.3 shown in the procedure. The rearranged equations are shown below:

$$\zeta = \frac{-\ln(PO/100)}{\sqrt{\pi^2 + \ln^2(PO/100)}} \quad (3.1)$$

$$\omega_n = \frac{\pi}{t_p \sqrt{1 - \zeta^2}} \quad (3.2)$$

Before solving for the damping ratio, the percent overshoot was first calculated using Equation 2.1 with the measured response parameters found using Figure 16.

$$PO = 100 \left(\frac{y_{max} - y_f}{y_f} \right) = 100 \left(\frac{0.4663 - 0.2578}{0.2578} \right) \rightarrow PO = 80.86\%$$

Now with the PO, the damping ratio and natural frequency could be solved for using the previously defined equations 3.1 and 3.2:

$$\zeta = \frac{-\ln(80.86/100)}{\sqrt{\pi^2 + \ln^2(80.86/100)}} \rightarrow \zeta = 0.0675$$

$$\omega_n = \frac{\pi}{1.342 \sqrt{1 - 0.0675^2}} \rightarrow \omega_n = 2.3463 \text{ rad/s}$$

III.7 Calculating PID Gains

Using the second-order system parameters found in Section III.6 and targeting a critically damped response ($\zeta = 1$) with a 5% settling time of 2 seconds ($\omega_n = 2 \text{ rad/s}$) and a zero position of $p_0 = 1$, the PID gains were calculated using equation 3.12-3.14 detailed in the lab workbook, shown below.

$$K_p = \frac{\omega_n^2 + 2\zeta\omega_n p_0 - c}{a} \quad (3.3)$$

$$K_i = \frac{\omega_n^2 p_0}{a} \quad (3.4)$$

$$K_d = \frac{2\zeta\omega_n + p_0 - b}{a} \quad (3.5)$$

The plant transfer function coefficients denoted as a, b, and c in the equations above were taken from the transfer function of the second-order approximation of the Quanser Aero Plant:

$$\frac{Y(s)}{R(s)} = \frac{y_f \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2(1 - y_f)} \quad (3.6)$$

where:

$$a = y_f \omega_n^2 \quad (3.7)$$

$$b = 2\zeta\omega_n \quad (3.8)$$

$$c = \omega_n^2(1 - y_f) \quad (3.9)$$

Substituting the plant coefficients into the gain equations and solving:

$$K_p = \frac{2^2 + 2(1)(2)1 - 2(1 - 0.2578)}{0.2578(2)^2} \rightarrow K_p = 3.4570$$

$$K_i = \frac{2^2(1)}{0.2578(2^2)} \rightarrow K_i = 3.1677$$

$$K_d = \frac{2(1)2 + 1 - 2(1)2}{0.2578(2)^2} \rightarrow K_d = 3.8800$$

Table 4: Calculated and Provided PID Gains

Section	Description	Symbol	Value	Unit
Section 4.2.1	Calculated Gains			
	Proportional Gain	K_p	3.4570	-
	Integral Gain	K_i	3.1677	-
	Derivative Gain	K_d	3.8800	-
	Provided Gains			
	Proportional Gain	K_p	3.3739	-
	Integral Gain	K_i	3.06	-
	Derivative Gain	K_d	3.7484	-

These gains are compared to the gains that were provided in the lab workbook in the analysis. The compensated Aero system response was then plotted by applying the workbook provided gains, shown above, to the system with a 0.3 radian square wave input. Figure 16 shows the resulting angle response and Figure 17 shows the voltage response.

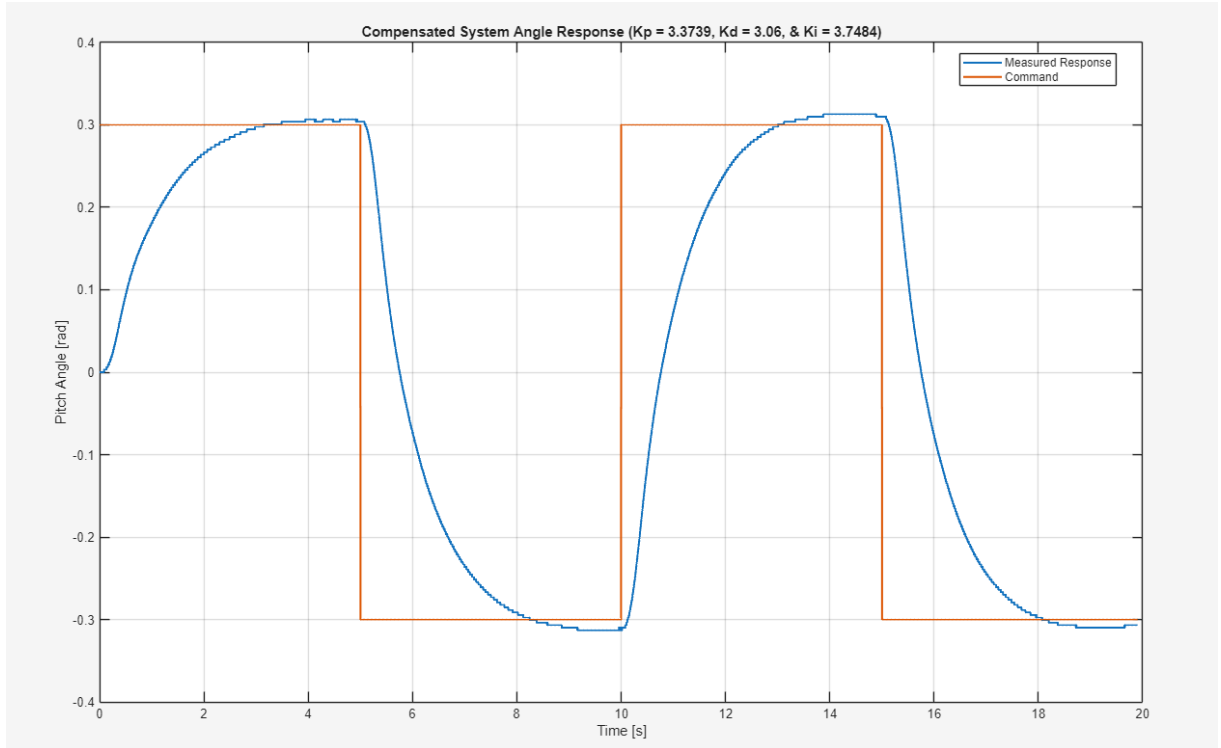


Figure 16: Compensated system pitch angle response using given PID gains ($K_p = 3.3739$, $K_i = 3.06$, $K_d = 3.7484$).

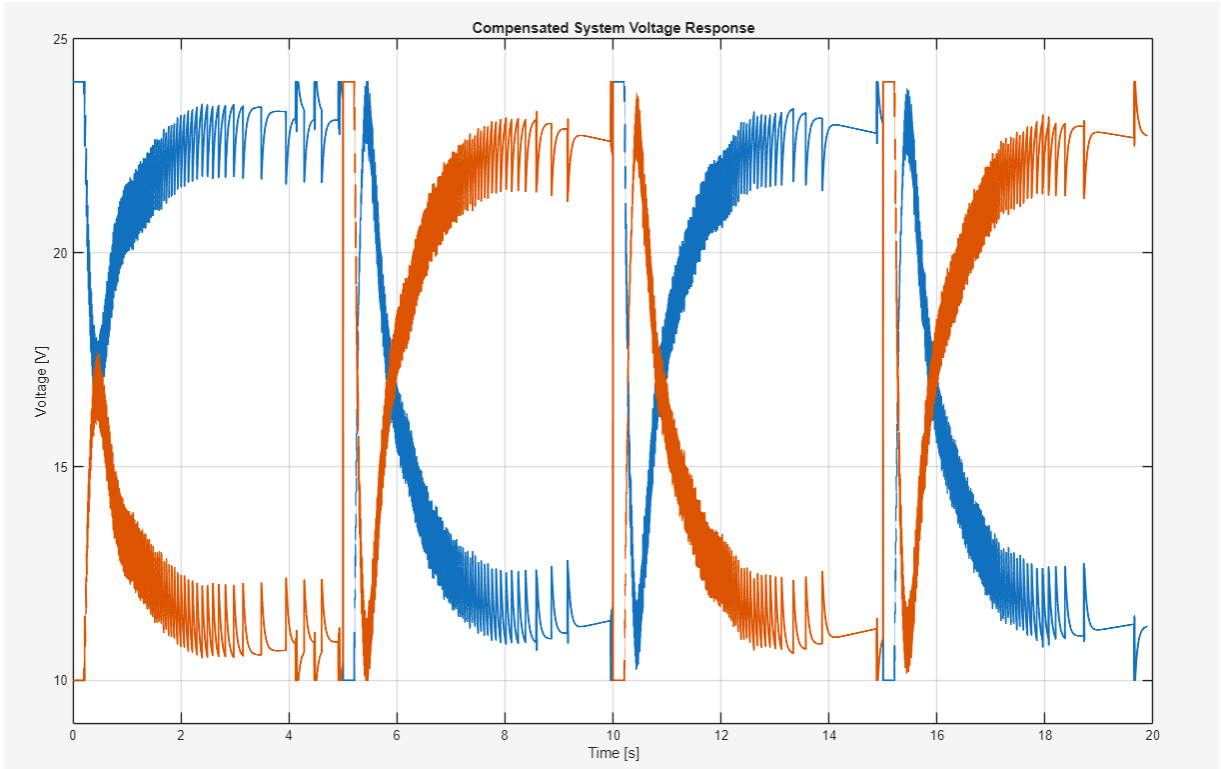


Figure 17: Compensated system voltage response using provided workbook PID gains.

IV. ANALYSIS

IV.1 Proportional Control

Proportional control alone is not practical for meeting a peak time under 1.1 seconds. A higher proportional gain does speed up the rise, but it also increases the overshoot and keeps the system oscillating. This is shown in figure 3 where each tested gain produced oscillations that never fully died out over the full 20 second run. There is no damping mechanism in a P controller, and no way to eliminate steady-state error, so you need the other terms.

IV.2 Derivative Control

Adding derivative control to the system helps to stabilize those large oscillations seen with the P controller. Figure 5 shows that as the derivative gain increases from 4 to 8, the system settles faster and overshoots less. Figure 7 shows what happens when the derivative gain is too small. The gain was set to 0.24 where it did almost nothing to the response, so it looks nearly identical to the proportional only graphs. A derivative gain of 2.4 was found to be the minimum value that kept the peak pitch within the ± 0.35 radian limit on the square wave input, as shown in figure 8.

IV.3 Integral Control

As seen in figure 9, adding integral control reduces the steady-state error that still exists with the PD controller. The integral term accumulates the error over time and continues to drive the system towards the reference until the error is zero. However, if the integral gain is increased too much the system may overshoot more and oscillate longer before settling. With that said, an integral gain value of five was found to bring the steady-state response within the required 3% band of 0.291 to 0.309 radians.

IV.4 Response Tuning

The tuning process started from the derivative and integral gains found in the previous sections and then the proportional control was increased until the peak time came in under 1.1 seconds. However, as the proportional gain was increased the overshoot also increased, so the derivative gain also had to increase to compensate. The integral gain was left at five since the steady-state was already within the required bounds. The selected final gains met all three required specifications.

IV.5 Second Order Approximation

Figure 15 compares the measured aero response and the second-order transfer function model. The two curves follow each other though the first peak, with a similar rise time and amplitude. After around six seconds, the measured response starts to deviate from the model with higher peaks and a slightly slower decay rate.

This deviation is expected given that the second-order model doesn't account for all of the physical dynamics present in the real system, including prop aero and motor inductance. These effects are small early on in the transient but cascade and become more apparent as the response progresses.

For the quantitative error analysis, the first peak amplitude differs by about 0.02 radians between the two curves and grows to about 0.04 radians by the time it reaches the second peak. Both curves settle to around 0.26 radians showing that the model captured the DC gain of the system. Despite the minor difference in the transient amplitude, the second approximation is sufficient to serve as a baseline for calculating the PID controller gains.

IV.6 Comparison of Tuning Methods

The two tuning methods resulted in two sets of PID controller gains that were not very close to each other. Iteratively tuning the response results with a proportional gain of 15, integral gain of 5, and a derivative gain of 4.5. The analytical method resulted in a proportional gain of 3.4570, integral gain of 3.17, and a derivative gain of 3.88, which was very close to the given workbook values of 3.3739, 3.06, and 3.7484. The largest gap between the two methods is with the proportional gain with 15 vs 3.46. This difference comes down to what the goal that each method was chasing.

The analytical design targets critical damping with a 2 second settling time while prioritizing a smooth response that doesn't overshoot over response time. The iterative tuning was driven by the 1.1 second peak time specification which required the proportional gain to be much higher to make the response react fast enough. The results reflected that directly.

The tuned system in figure 12 hits all three specifications while the analytical system in figure 16 only hits two of the three. The peak time was measured to be 3.15 seconds, which is well over the 1.1 second limit, but that isn't a failure of the tuning method. The controller was designed for critically damping not a fast response which is shown directly in figure 16. When a smoother settling response is preferred over speed, the analytical gains would be the better choice.

V. HOME ANALYSIS

5.1 Frequency Analysis

Symbol	Description	Value
y_f	Steady-state final value	0.2546 rad
ω_n	Natural frequency of plant	2.213 rad/s
ζ	Damping ratio of plant	0.0227
$a = y_f \omega_n^2$	Plant gain numerator	1.2469
$b = 2\zeta \omega_n^2$	Plant 1 st -order coeff. denominator	0.1005
$c = \omega_n^2(1 - y_f)$	Plant 0 th -order coeff. denominator	3.6505
k_p	Proportional gain	3.4570
k_i	Integral gain	3.1677
k_d	Derivative gain	3.8800

The second-order closed-loop approximation is determined using the following:

$$\frac{Y(s)}{R(s)} = \frac{y_f \cdot \omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2(1 - y_f)}$$

Plugging in $y_f = 0.2546$, $\omega_n = 2.213$, $\zeta = 0.0227$:

$$\frac{Y(s)}{R(s)} = \frac{(0.2546 \cdot 2.213^2)}{s^2 + 2(0.0227)(2.213)s + (2.213)^2(1 - (0.2546))} = \frac{1.2469}{s^2 + 0.1005s + 3.6505}$$

This expression obtained can be used as P(s) for frequency analysis

$$P(s) = \frac{1.2469}{s^2 + 0.1005s + 3.6505}$$

From modifying Eq. 1.15 in the workbook, we obtain the following expression:

$$C(s) = k_p + \frac{k_i}{s} + k_d \cdot \frac{Ns}{(s + N)}$$

After further manipulation we end up with the following expression:

$$C(s) = \frac{(k_d \cdot N + k_p)s^2 + (k_p \cdot N + k_i)s + k_i \cdot N}{s(s + N)}$$

Plugging in $k_p = 3.4570$, $k_i = 3.1677$, $k_d = 3.8800$, $N = 20$:

$$C(s) = \frac{(3.8800 \cdot 20 + 3.4570)s^2 + (3.4570 \cdot 20 + 3.1677)s + (3.1677 \cdot 20)}{s(s + 20)}$$

$$= \frac{80.057s^2 + 72.267s + 63.354}{s^2 + 20s}$$

From the relationship $L(s) = C(s) \cdot P(s)$:

$$L(s) = ((80.057s^2 + 72.267s + 63.354) \cdot 1.2469) / ((s^2 + 20s) \cdot (s^2 + 0.1005s + 3.6505))$$

$$L(s) = (99.847s^2 + 90.126s + 79.023) / (s^4 + 20.101s^3 + 5.651s^2 + 73.01s)$$

Unity-feedback of the closed-loop transfer function is:

$$T(s) = \frac{L(s)}{1 + L(s)}$$

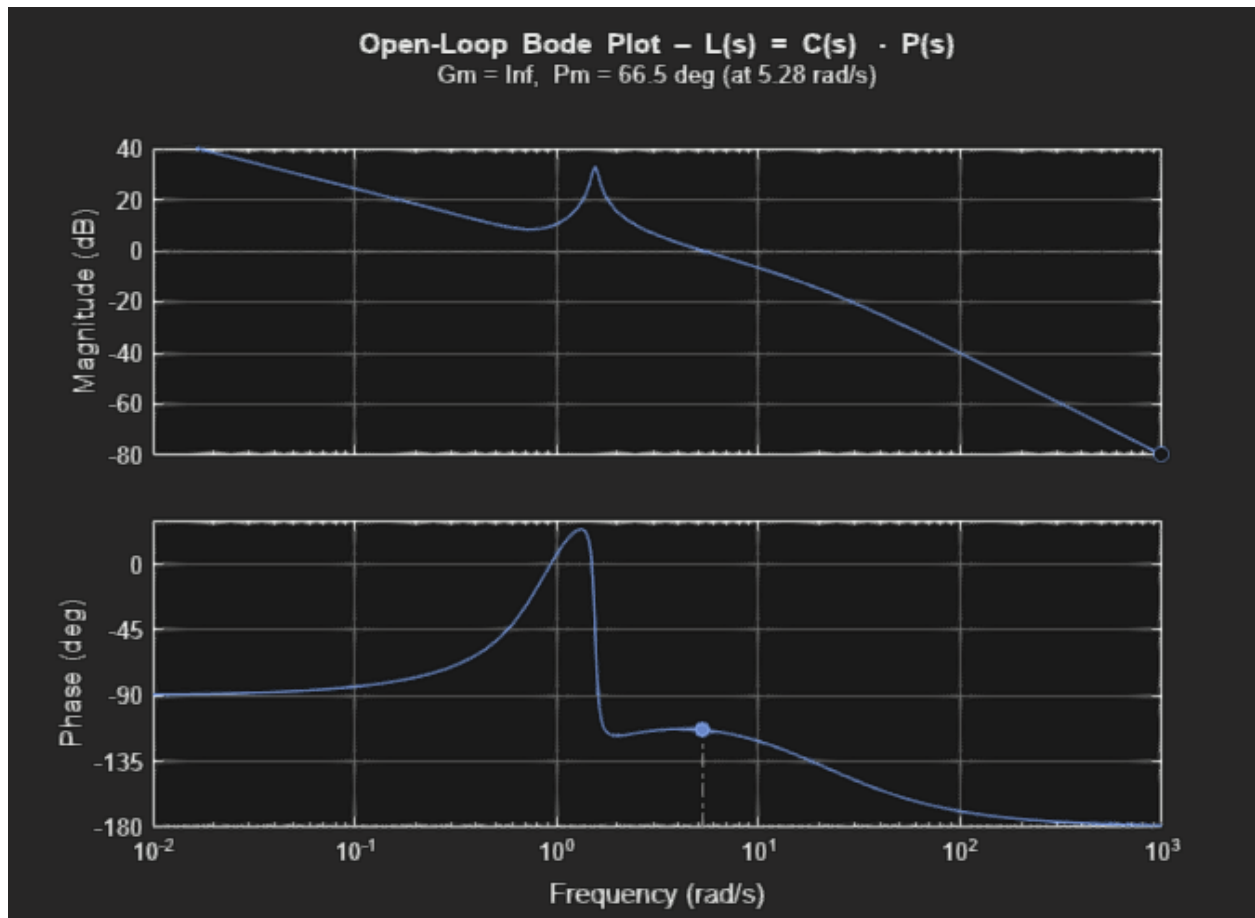


Figure 18: Open-Loop Bode Plot Diagram

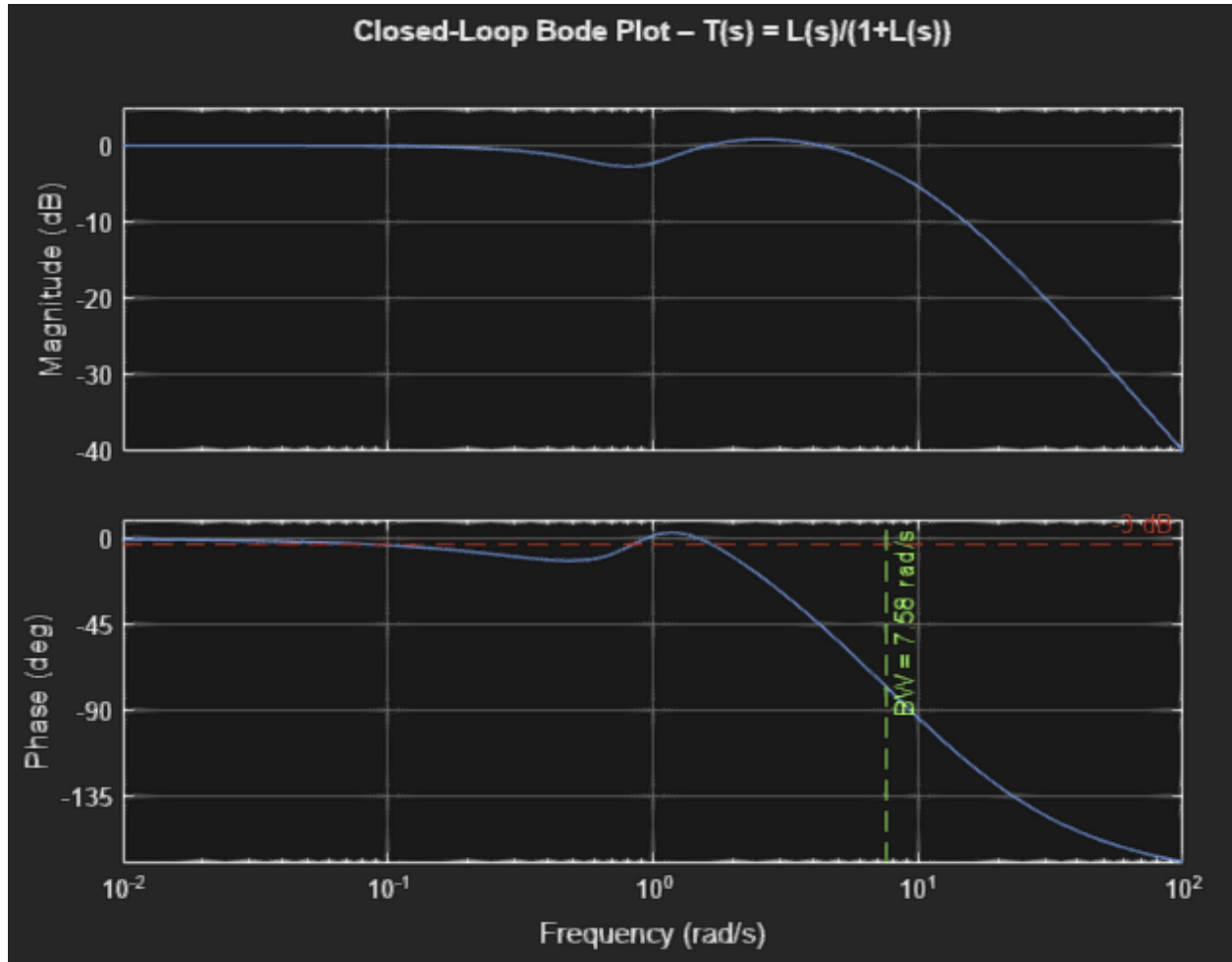


Figure 19: Closed-Loop Bode Plot Diagram

From interpretation of MATLAB output and these plots, we see that phase margin (66.39°) is well above the 45° guideline, the gain margin is infinite as it never reaches -180° , gain crossover frequency is 5.48 rad/s and the bandwidth (BW) is at 7.58 rad/s.

Open-Loop: $P(s)$ has natural frequency of 2.213 rad/s and damping ratio 0.0227 which is nearly undamped. The open-loop poles were determined to be at $-0.050 \pm 1.911j$ rad/s, so without compensation, any unity-feedback loop would lead to a sustained oscillation.

Closed-Loop: $T(s)$ achieves a bandwidth of 7.58 rad/s which is more than 3.4 times the uncompensated $P(s)$ natural frequency of 2.213 rad/s. The unity DC gain confirms zero steady-state error with a mild peak of 0.74 dB at 3 rad/s indicating slight underdamping which is consistent with the nearly critically-damped PID target of the design.

Effects of designed PID controller: The PID adding an integrating pole ki/s forces the open-loop gain to infinity and ensures zero steady-state error to a step command. The derivative terms creates a phase lead that helps offset plant lag. This compensated open-loop $L(s)$ achieves phase margin (66.39°) which is well above the 45° guideline.

5.2 Frequency Analysis

Putting the plant $P(s)$ into state-space form with states $x_1 = \theta, x_2 = \dot{\theta}$

$$x_1 = x_2$$

$$\dot{x}_2 = -c \cdot x_1 - b \cdot x_2 + a \cdot u = -3.6505 \cdot x_1 - 0.1005 \cdot x_2 + 1.2469 \cdot u$$

$$y = x_1 = \theta$$

$$\dot{x} = A \cdot x + B \cdot u, y = C \cdot x$$

System Matrices are the following:

$$A = [0, 1; -3.6505, -0.1005] \quad B = [0; 1.2469] \quad C = [1, 0] \quad D = 0$$

Controllability Matrix is:

$$W_c = [B, AB]$$

$$A \cdot B = [0, 1; -3.6505, -0.1005] \cdot [0; 1.2469] = [1.2469; -0.1253]$$

Determinant is:

$$\det(W_c) = (0)(-0.1253) - (1.2469)(1.2469) = -1.555 \neq 0$$

$$\text{rank}(W_c) = n = 2$$

Because $\text{rank} = n$ the system is fully controllable by definition. Any state that is desired can be reached from any given initial condition through an appropriate voltage input. The differential thruster voltage can control both the pitch angle and pitch rate.

Observability Matrix is:

$$W_o = [C; CA]$$

$$C \cdot A = [1, 0] \cdot [0, 1; -3.6505, -0.1005] = [0, 1]$$

$\text{Rank}(W_o) = n = 2$ so the system is fully observable by definition. States θ and $\dot{\theta}$ can be built from encoder output $y = \theta$. Internal dynamics are not hidden from the sensor.

Quadratic Cost Functional Definition of LQR:

$$J = \int_0^\infty (x^T Q x + u^T R u) dt$$

$$u = -K \cdot x$$

$$Q = \text{diag}(10, 1), R = 1$$

$$A^T P + P A - P B R^{-1} B^T P + Q = 0$$

$$K = R^{-1}B^T P$$

Solving using matrices above:

$$K = [1.3817, 1.7146]$$

$$A_{cl} = A - BK$$

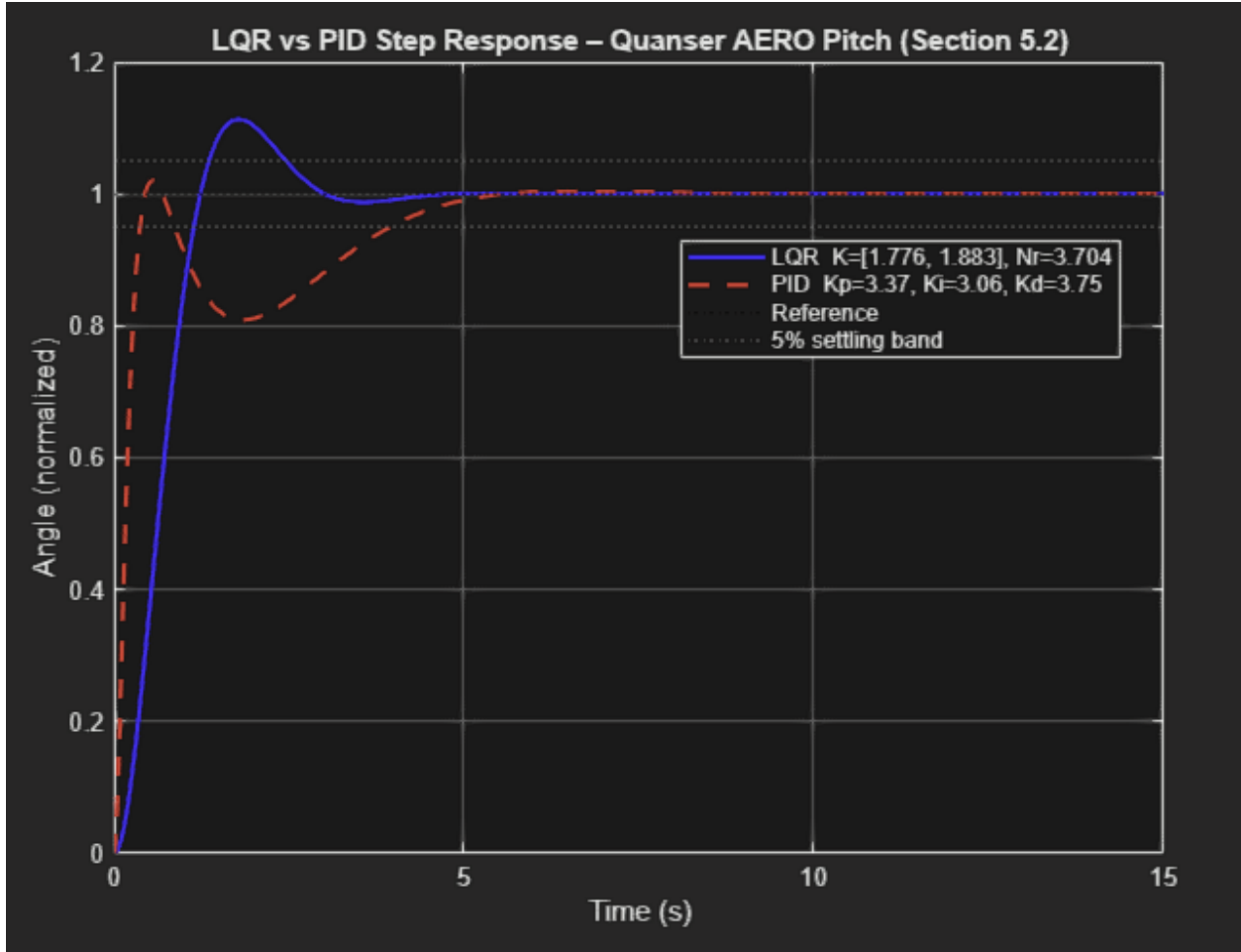
$$\lambda_{1,2} = -1.119 \pm 2.030j$$

Computing feedforward gain N_r to achieve unity feedback gain:

$$N_r = \frac{1}{dcgain(A_{cl}, B, C, 0)} = 4.310$$

Complete Control Law with Reference Tracking:

$$u = N_r * r - Kx = 4.310 * r - [1.3817\theta + 1.7146\dot{\theta}]$$



Motor voltage can be related to control input:

$$V_{motor} = V_{level} \pm V_{gain} * u = 17 \pm 7u$$

Because linear operating range is 10-24 V, we can determine maximum allowable control input:

$$u_{max} = \frac{24 - 17}{7} = 1$$

$$u(0) = N_r * r - K * 0 = 4.310 * 0.3 = 1.293$$

After Substitution:

$$V_{motor} = 17 + 7 * 1.293 = 26.05 \text{ V} > 24 \text{ V}$$

Initial LQR exceeds $u_{max}=1$ by 29.3% which leads to motor saturation at the beginning of each step. This means that the linear model is no longer valid, actual saturation is lower than predicted which causes a transient error until the system goes back into the linear motor range. The PID controller was tuned and verified to remain within this linear motor range. The tuned PID controller is a more practical solution compared to the LQR as you would need to increase R to

lower $u_{\max}$ as it is currently over by 29.3%. Increasing R would reduce K and N_r which would trade speed of the response for actuator feasibility.

VI. CONCLUSIONS

The PID controller is a mechanism that is meant for correcting feedback errors by calculating the difference between the desired and measured value and alters the system's output based on that difference. When the proportional controller was implemented to the Quanser motor system, the pitch angle responses increased to the input response. However, the controller also increased the overshoot and produced more oscillations to the system, making the motors underdamped. When the derivative controller was added to the diagram, the oscillations were mitigated to the point of stabilizing with and the overshoot also reduced but still provided some difference between the desired and measured responses. The integral controller was able to reduce the remaining steady-state error and tune the output towards the command signal. This allowed for stabilization towards a desired value. During the response tuning, the proportional gain was altered to 15 to set the peak time within 1.1 seconds while the derivative gain was set to 4.5 to remove the excess oscillations. The integral gain set to 5 allowed for a near-zero steady-state error, allowing for the Aero to meet its design and test criteria. The approximation of the second order system showed some skewness with the measured response but the quantity of the deviation was very minimal and insignificant, allowing the values to be considered valid for the controller gains. When determining the K_p , K_i , and K_d values through the analytical method, the results found did not fully match with the products of the responsive tuning, mainly the proportional gains. The analytical method also produced a peak time that went outside the defined limit. This however, didn't label the approximation results as invalid as the controller was able to provide an acceptable steady-state error while having no excessive oscillations while stabilizing. The overall success in the validity of the experiments evidenced that the Quanser lab provided an excellent implementation towards PID controllers in feedback loops.

V. REFERENCES

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